

Design Analysis & UX/UI Design

Problem Framing, Brand Architecture, Wireframes & Guidelines

PROJECT	TRACE — Understanding Microplastics. Taking Action.
TRACK	Graphic Design
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SOURCE MATERIAL	Project Proposal.pdf, Trace Presentation.pdf (Brand Identity & Strategy Presentation)

Understanding Microplastics. Taking Action.

3.1 Problem Statement & Objectives

Eco-conscious young adults in fast-paced urban environments need a transparent, science-backed tool to detect and reduce their daily biological exposure to microplastics. They are currently paralyzed by the sub-visible nature of the threat, pervasive corporate greenwashing, and a lack of budget-friendly solutions that fit their daily lifestyle.

Design objective: resolve this problem visually and experientially — not by building the scanning technology itself, but by designing a brand and interface that make the invisible threat legible, personal, and actionable within a single, coherent design system.

3.2 Brand & UX Design Architecture

In place of a software architecture diagram (not applicable — no coded system exists), the diagram below describes how the design system's components connect: foundational identity decisions feed every downstream application.

LAYER	COMPONENT	FEEDS INTO
1. Foundation	Brand summary, problem statement, persona, SWOT	Logo & concept direction
2. Identity core	Logo & concept, color system, typography, moodboard	Every application below
3. Applications	Mobile app, website, social media, merchandise, business card	Brand guidelines manual (Document 06)

3.3 Wireframes & Mockups

MOBILE APP

Low-fidelity sketch wireframes were produced first for the four core screens — Home/Learn, Impact, Take Action, and Community — establishing layout and content hierarchy before visual styling. These were then resolved into a high-fidelity four-step flow: **01 Learn → 02 Scan → 03 Take Action → 04 Community**, plus two supporting features: a Body Burden Estimator (risk broken down by lungs, water, food, and lifestyle) and a Scan screen supporting four capture modes (Product, Clothing, Food/Drink, Water).

BUSINESS CARD

A full wireframe specification defines the front side (5mm rounded corners, logo placement, tagline zone, background wave layers, slogan) and back side (logo placement, text-field zone, and a dedicated

QR-code box alongside phone/email/web contact-data blocks).

WEBSITE

A responsive homepage design was produced for desktop and mobile: a hero section (“Science that protects our future”), a proof-point stats bar, and a “What We Do” services grid (Sample Analysis, Accurate Detection, Data & Insights, Environmental Impact).

DESIGN MAXIM

“Type is structure. Clarity is performance.” — the typography rationale from the Trace Presentation, carried forward as the system's working principle.

3.4 UI/UX Guidelines

COLOR PALETTE

Deep Teal	Teal	Aqua	Pale Aqua	Mist	Frost White
#0F4C5C	#2E6E7E	#6FA8A8	#A8C5C1	#DDEAE8	#F4FAFB

Deep teal and teal anchor headlines and primary UI chrome; aqua and pale aqua carry secondary accents and data visualizations; mist and frost white are reserved for backgrounds and negative space, keeping the “clinical clarity” tone the brand strategy calls for.

TYPOGRAPHY

Primary display face: **Clash Display (Heavy)** for headlines — confident and geometric for impact. Secondary face: **Inter**, Semibold for UI labels/precision touchpoints and Regular for body copy, keeping the system readable, calm, and functional. (Montserrat Bold/SemiBold/Regular appears in the early moodboard exploration; Clash Display + Inter is the typography system carried into final application design.)

ICONOGRAPHY

A consistent line-icon set — microscope, DNA helix, droplet, leaf, shield — supports the four brand pillars (Understand, Research, Protect, Together) at a uniform stroke weight across app, web, and print touchpoints.

ACCESSIBILITY

- Dark navy/teal text on frost-white or mist backgrounds, and white text on deep-teal or navy backgrounds, are the only approved pairs — both validated against WCAG AA contrast minimums.
- Minimum body copy size of 14px on mobile UI; display type never drops below 18px in-app.